

Prince Ude Backend and Infrastructure Engineer

ude.prncedev@icloud.com • Lagos, Nigeria • ude-p.github.io • in/prince-ude

Summary

Backend and Infrastructure Engineer specializing in Go, Kubernetes, and realtime backend systems. Shipped a multiplayer backend serving 600k+ users, telemetry infrastructure processing location updates from 3.5k+ concurrently online vehicles, and a Kubernetes operator that enables application deployment through custom resources and automated reconciliation.

Skills

- **Languages:** Go | SQL | Bash
- **Infrastructure & Ops:** K8s | k3s | Helm | controller-runtime | CRDs | Docker | Linux | Docker Compose | GitHub Actions | Caddy
- **Databases & Messaging:** PostgreSQL | TimescaleDB | Redis | NATS
- **Networking:** gorilla/websocket | gRPC | Protocol Buffers | TCP/WebSocket
- **Observability & Performance:** OpenTelemetry | OTel Collector | Grafana Cloud | Pyroscope | Pprof

Experience

Backend and Infrastructure Engineer02/2024 – Present
WHOT Studios 

Abuja, Nigeria

Gaming technology company building multiplayer mobile experiences serving hundreds of thousands of players.

- Built realtime networking infrastructure powering gameplay systems used by 600k+ players, implementing room state synchronization and websocket communication services in Go.
- Built authoritative game rooms supporting 500–1000 concurrent players per node with p99 tick latency of approximately 42ms.
- Profiled and optimized websocket fan-out bottlenecks affecting realtime game simulations running at 60 ticks per second, improving tick stability and reducing broadcast pressure under load.
- Migrated production workloads from single-server deployments to a self-managed k3s cluster, enabling horizontal scaling and standardized deployment workflows.
- Managed cluster deployments using Helm, GitHub Actions, ingress configuration, service networking, and rollout automation.
- Built asynchronous backend workflows using NATS for gameplay events and internal service messaging
- Implemented observability across backend services using OpenTelemetry, OTel Collector, Grafana Cloud, and runtime profiling.

Systems Reliability Engineer08/2022 – 02/2024
Gtrac Services Ltd

Lagos, Nigeria

Fleet management and GPS tracking company providing realtime telemetry and operational visibility solutions.

- Maintained telemetry ingestion infrastructure processing realtime location updates from 3.5k+ concurrently online vehicles.
- Built and maintained backend services supporting GPS tracking, telemetry ingestion, geofencing, alerts, and fleet reporting systems.
- Designed TimescaleDB and PostgreSQL ingestion pipelines handling high volume telemetry and time series workloads.
- Managed Linux-based production systems running containerized backend services, Caddy reverse proxy infrastructure, and Docker networking.
- Automated deployments using GitHub Actions and remote Linux rollout scripts.
- Implemented monitoring and observability using OpenTelemetry and Grafana Cloud.

Projects

AppDeployment Operator - Internal Kubernetes Deployment Platform

WHOT Studios

- Built a Kubernetes operator in Go using controller-runtime and a custom AppDeployment CRD, enabling developers to deploy services through a simplified declarative interface instead of managing Kubernetes resources.
- Automated Deployment, Service, and ConfigMap lifecycle management, rollout health checks, rollback handling, and CI/CD integration via GitHub Actions.

Cash.io - Multiplayer Realtime Infrastructure

WHOT Studios (playcash.io)

- Designed room lifecycle management, state replication, and snapshot synchronization mechanisms for authoritative multiplayer game servers in Go.
- Reduced bandwidth and allocation overhead via protobuf-based snapshot serialization; profiled goroutine pressure, websocket backlog buildup, and serialization hotspots with Pyroscope and pprof.

Routenex - Fleet Tracking SaaS

routenex.io

- Built GPS tracking and telemetry ingestion backend handling millions of writes into TimescaleDB/PostgreSQL and Redis-backed realtime systems.
- Designed and operated a self-hosted k3s platform with Helm, GitHub Actions, OpenTelemetry, and Grafana Cloud.